REV_DESIGN OF MICROCONTROLLER BASED ROBOTICS APPLICATION

Design of Microcontroller based Robotics Application

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code
4332

Credits
4

Periods
5/week

Course Details

Title: Design of Microcontroller based Robotics Application

Department: Common

Revision: REV_Design of Microcontroller based Robotics Application

Pattern: Theory / Practical

Complies with official SITTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 4332 — Design of Microcontroller based Robotics Application**. Use the official SITTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

Official SITTTR Revision 2026 syllabus reference. This page is a study handbook, not a substitute for the latest official notification or examination instruction.

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Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

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Navigate handbook

OBJECTIVES

Course Objectives and Outcomes

Objectives

1. Provide deep technical understanding of Design of Microcontroller based Robotics Application.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

CO-PO Mapping

CO1 → PO1: 3. CO2 → PO2: 3. CO3 → PO3: 3. CO4 → PO3: 3.

MODULE 1

Core Principles

Outcomes

Master foundational terminology and core definitions of Design of Microcontroller based Robotics Application.

Study

Detailed analysis of core principles and theoretical foundations.

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MODULE 2

Applied Methods

Outcomes

Apply standard calculation techniques and operational procedures.

Study

Numerical problem-solving techniques and practical workflows.

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MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

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MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

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MODEL PAPER

Practice Model Question Paper

Time: 3 Hours | **Max Marks:** 75

Part A (2 marks each)

1. Define core concepts of Design of Microcontroller based Robotics Application.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

ANSWERS

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.